

### SOLUCIONES TRABAJO PRACTICO N° 5

$$1) a) A = \int_0^{\theta_1} \rho_1^2 d\theta + \int_{\theta_1}^{\pi} \rho_2^2 d\theta$$

$$b) A = \frac{1}{2} \int_0^{\theta_1} \rho_1^2 d\theta + \frac{1}{2} \int_{\theta_1}^{\pi} (\rho_1^2 - \rho_2^2) + \frac{1}{2} \int_{\pi}^{3\pi/2} \rho_2^2 d\theta$$

$$2) A = \int_0^{\pi/3} \left(\frac{3}{2}\right)^2 d\theta + \int_{\pi/3}^{\pi} (1 + \cos(\theta))^2 d\theta = \frac{7}{4}\pi - \frac{9}{8}\sqrt{3}$$

$$3) A = 2 \int_{\pi/12}^{5\pi/12} \left( (2\sin(2\theta))^2 - 1^2 \right) d\theta = \frac{2\pi}{3} + \sqrt{3}$$

$$4) A = \int_0^{\pi/6} (\cos^2 \theta - \sin^2(2\theta)) d\theta = \frac{3\sqrt{3}}{16}$$

$$5) a) A = \int_1^{\sqrt{3}} (7t^2 + 4t) \frac{1}{t} dt = 3 + 4\sqrt{3}$$

$$b) A = \int_0^1 t e^{7t} dt = \frac{e^7}{7} - \frac{e^7}{49} + \frac{1}{49}$$

$$6) A = - \int_{2\pi/3}^0 6 \sin^2 \theta d\theta = 2\pi + \frac{3}{4}\sqrt{3}$$